Coding & Solution

Date	01 July 2026
Project Name	OptiCrop
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code for OptiCrop and the overall solution. The submission should demonstrate working functionality for crop recommendation and environmental suitability assessment, clean code practices, and alignment with the defined requirements across all three usage scenarios.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/koppojupravallika/OptiCrop
Programming Language(s)	Python
Framework(s) Used	Flask / Streamlit, Scikit-learn, Pandas, NumPy
Key Features Implemented	Crop recommendation based on N, P, K, temperature, humidity, pH and rainfall inputs; crop suitability and environmental assessment; data visualization for research and policy insights
Pending / Incomplete Features	• User authentication (Login/Registration). • Prediction history and report download. • Real-time weather API integration. • Admin dashboard for monitoring predictions and users.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project follows a modular architecture with separate frontend and backend components. Python (Flask) is used for backend processing, while HTML, CSS, and JavaScript are used for the user interface. The application integrates a Random Forest machine learning model to recommend suitable crops based on soil nutrients and environmental conditions. The code is organized for readability and maintainability, with meaningful naming conventions and reusable modules.